

JAY SHAH

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SUMMARY

Entry-level Software Engineer with M.S. Computer Science and hands-on experience designing, building, and deploying **full-stack applications**, **REST APIs**, and **backend services**. Proficient in **Python**, **Java**, **JavaScript/TypeScript**, and **SQL**, with experience in **cloud deployment**, **CI/CD pipelines**, and **Agile development**. Strong foundation in data structures, algorithms, system design, and writing clean, well-tested, maintainable code.

WORK EXPERIENCE

University of Texas, Arlington

Aug 2024 – May 2026

Graduate Assistant — Software Engineering and AI Systems

Arlington, TX

- 60+ software projects had no shared development standard – code was inconsistent, failures were caught manually, and there was no automated testing or CI process to validate submissions before review.
- Built **Python** automation tooling with clean, modular code; wrote **SQL** validation scripts; implemented unit and integration test suites; created monitoring dashboards in **Plotly**; documented all systems and processes following **SDLC** best practices for faculty and cross-functional research teams.
- 30% fewer failures**, **25% faster** review cycles – well-structured, documented software that any team member could understand, test, and extend without needing additional support.

Tata Consultancy Services

Jan 2024 – May 2024

Software Development Engineer Intern

Gandhinagar, India

- Production backend services processing **500K+ records** had broken integrations between system components – other teams were blocked, a silent bug had been corrupting outputs for weeks, and there was no monitoring to catch failures automatically.
- Designed and built **Python + FastAPI** REST backend services; wrote **SQL** data transformation layers; implemented input validation and error handling; added monitoring and logging; deployed services on **AWS (S3, EC2)**; followed **Agile Scrum** sprint workflows with code reviews and documentation.
- 2 features shipped** to production users, system **30% more reliable**, delivery **15% faster** – clean, well-documented code that unblocked three dependent teams and eliminated an entire category of runtime failures.

Celebal Technologies

May 2023 – Sept 2023

Software Development Engineer Intern

Jaipur, India

- Three client teams needed custom software to replace manual workflows – existing systems were fragile, undocumented, and couldn't be adapted to meet their specific business requirements without breaking something else.
- Developed **Java** and **Python** backend services; designed **SQL** database schemas; built frontend dashboards in **JavaScript**; wrote unit tests; participated in code reviews; delivered user documentation and technical specs following **Agile** practices across three different client environments.
- Manual workload reduced **25%**, data accuracy improved **18%** – tailored, well-tested software delivered on schedule to three different client stacks, each with unique requirements.

PROJECTS

Full-Stack Web Application with REST API · React · TypeScript · FastAPI · Python · PostgreSQL · Docker · AWS · GitHub Actions

Dec 2025 – Mar 2026

- Teams had no unified platform to access and act on their data – information was scattered across tools, there was no shared API, and every update required a manual process that engineers avoided making.
- Built every layer end-to-end: **React + TypeScript** frontend with responsive UI, **FastAPI** backend with **REST API** endpoints, **PostgreSQL** database with optimized schema design, **Docker** containerization, **AWS** deployment, and a **GitHub Actions** CI/CD pipeline with automated testing and safe rollouts.
- Query performance **35% faster**, zero-downtime deployments, system handled **300K+ requests** reliably – engineers went from avoiding releases to shipping confidently every sprint.

Scalable Microservices Backend System · Python · Java · FastAPI · Spring Boot · PostgreSQL · Redis · Docker · Kubernetes · AWS

Apr – Jul 2025

- A monolithic backend was becoming a bottleneck – **12K+ concurrent users** caused timeouts, a single deployment failure took everything down, and the codebase was too tightly coupled for any team to work on independently.
- Redesigned as a **microservices architecture** using **Python (FastAPI)** and **Java (Spring Boot)**; implemented **Redis** caching for performance; containerized with **Docker** and orchestrated with **Kubernetes** on **AWS**; added structured logging, health checks, and automated integration tests across all service boundaries.
- System throughput improved **40%**, independent deployments per service, **99.9% uptime** – teams could develop, test, and deploy their services without touching or breaking anything else.

AI-Powered Feature for End Users · Python · LangChain · OpenAI APIs · FastAPI · React · PostgreSQL · Docker · GitHub Actions

Mar – May 2024

- Users had to manually search through thousands of records to find answers – the process was slow, frustrating, and a significant amount of the engineering team's time was spent answering repetitive data questions.
- Designed and built an **AI-powered search and query feature** end-to-end: **LangChain + OpenAI API** backend for natural language understanding, **FastAPI** REST service, **PostgreSQL** for structured data retrieval, **React** UI, **Docker** deployment, and full **CI/CD** with automated tests and monitoring via **GitHub Actions**.
- Manual query workload reduced **40%**, feature handled **1M+ requests** reliably – users got instant answers in plain English and engineering time shifted back to building new features.

EDUCATION

University of Texas, Arlington

Aug 2024 – May 2026

M.S. Computer Science **GPA: 3.90/4.00** · Distributed Systems · Software Engineering · Cloud Computing · ML

Arlington, TX

Charusat University

Oct 2020 – May 2024

B.S. Computer Engineering **GPA: 3.78/4.00** · Data Structures · Database Systems · Software Engineering · OOP

India

SKILLS

Languages: Python (advanced) · Java (proficient) · JavaScript / TypeScript (proficient) · SQL (advanced) · C++ (intermediate) · Bash

Backend: FastAPI · Flask · Spring Boot · REST API design · GraphQL (familiar) · gRPC (familiar) · microservices · event-driven architecture

Frontend: React · TypeScript · HTML · CSS · responsive design · component architecture · state management

Databases: PostgreSQL · MySQL · MongoDB · Redis · DynamoDB · SQL schema design · query optimization · ORM (SQLAlchemy)

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS) · GCP (familiar) · Azure (familiar) · Docker · Kubernetes · Terraform (familiar)

CI/CD & Testing: GitHub Actions · Jenkins (familiar) · unit testing · integration testing · pytest · JUnit · test-driven development

AI & ML Integration: LangChain · OpenAI APIs · PyTorch · scikit-learn · LLM integration · RAG · prompt engineering

System Design: Distributed systems · scalability patterns · caching · load balancing · message queues (Kafka, SQS) · API design

Process: Agile · Scrum · Git · code review · SDLC · technical documentation · pair programming